Final Capstone Project

Project Title

SDG 13: Climate Change

TerraSense AI – Sustainability Assistant Chatbot

Project Objective & Description

- **Objective:** The objective of TerraSense AI is to educate users about environmental sustainability and encourage eco-friendly habits through an interactive chatbot experience.
- **Description:** TerraSense AI is a sustainability-focused chatbot developed using Botpress. The chatbot provides users with information related to climate change, energy conservation, water conservation, waste management, and sustainable lifestyles.
  The chatbot acts as a digital sustainability assistant that helps users Understand environmental challenges and learn practical solutions that can be implemented in daily life.
  The project supports United Nations Sustainable Development Goal (SDG) 13: Climate Action.
- **Purpose:** The purpose of this project is to increase environmental awareness and promote sustainable behavior among individuals.
  Many people are aware of environmental issues but lack practical guidance on how to contribute to climate action. TerraSense AI bridges this gap by providing easy-to-understand information and actionable recommendations..
- **Use Cases:**

- Climate Change Awareness: Users can learn about the causes, effects, and solutions related to climate change.
- Energy Conservation: Users receive practical suggestions for reducing energy consumption at home, college, and during transportation.
- Water Conservation: Users learn methods to save water and understand the importance of conserving freshwater resources.
- Waste Management: Users receive guidance on waste segregation, recycling, reuse, and reducing plastic pollution.
- Sustainable Lifestyle: Users explore eco-friendly habits, sustainable shopping practices, and green transportation options.

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Technologies Used

- Botpress: Botpress was used as the primary chatbot development platform. It provided a visual workflow builder that enabled the creation of interactive conversation flows, topic navigation, and user interaction without extensive coding.
- Conversational AI: Conversational AI concepts were used to design natural and user-friendly interactions. The chatbot guides users through sustainability topics and provides structured information in a conversational format.
- Workflow Automation: Botpress workflow nodes, transitions, and multiple-choice interactions were used to create a smooth user journey. This helped users navigate different sustainability topics efficiently.
- GitHub: GitHub was used for project documentation and version control. The project repository stores project details, screenshots, and related files, making the work accessible and organized.
- Microsoft Word and PDF: Microsoft Word was used to prepare the project documentation, and the final report was converted into PDF format for submission and sharing.
- Sustainability Knowledge Content: Environmental and sustainability-related content was researched and organized to provide educational information on climate change, energy conservation, water conservation, waste management, and sustainable lifestyles.
- SDG Alignment : This project supports SDG 13 (Climate Change) by promoting environmental awareness and sustainable behaviour through an interactive chatbot.

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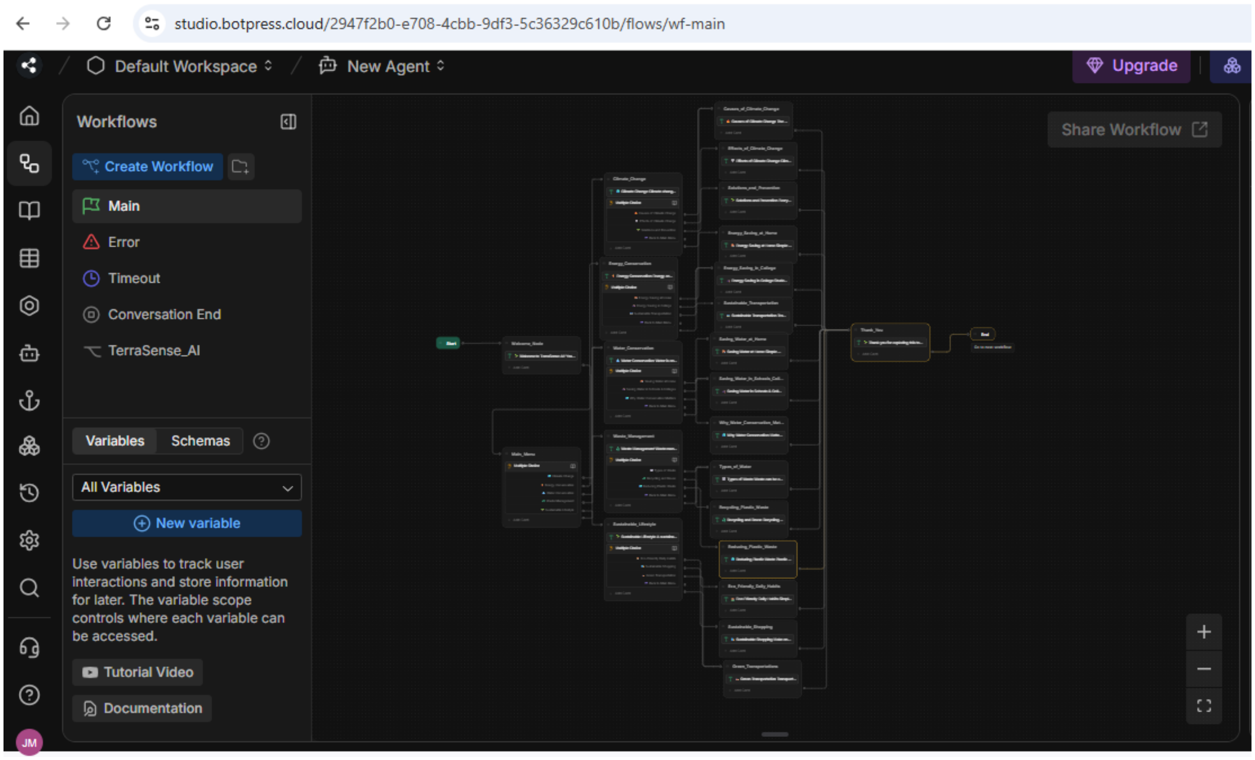
Screenshot and Explanation

User Input

The user interacts with TerraSense AI by selecting a sustainability topic from the available options. The input represents the user's area of interest, allowing the chatbot to provide relevant environmental information and guidance.



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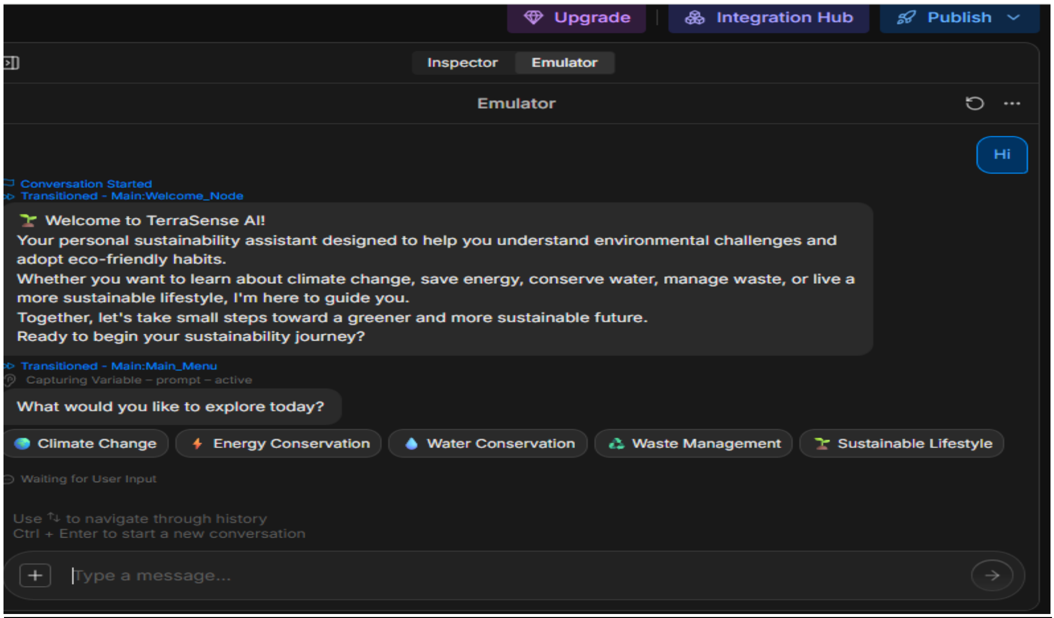
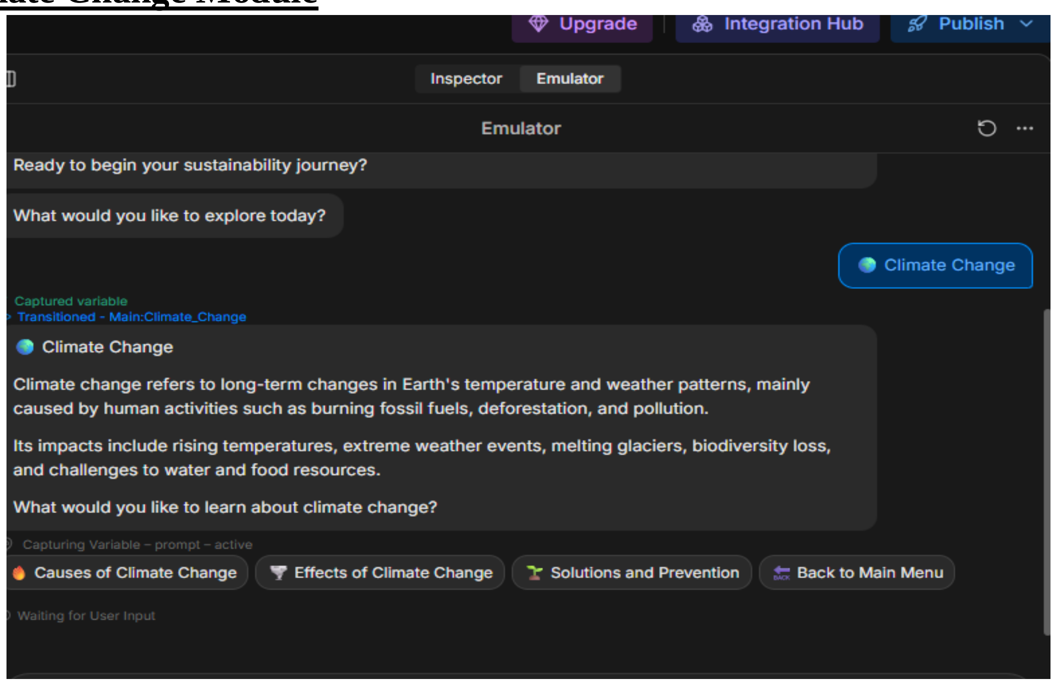
Chatbot Output

TerraSense AI processes the user's selection and provides informative responses, sustainability recommendations, and educational content related to the chosen topic. The chatbot aims to increase environmental awareness and encourage sustainable practices.

- **Welcome Screen**



- **Climate Change Module**



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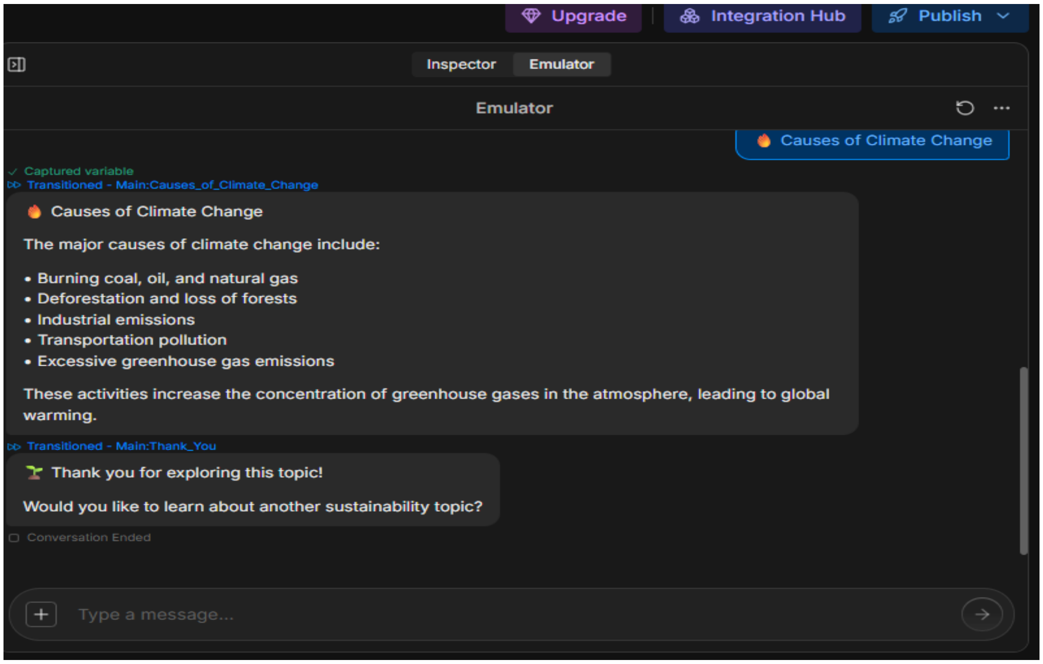
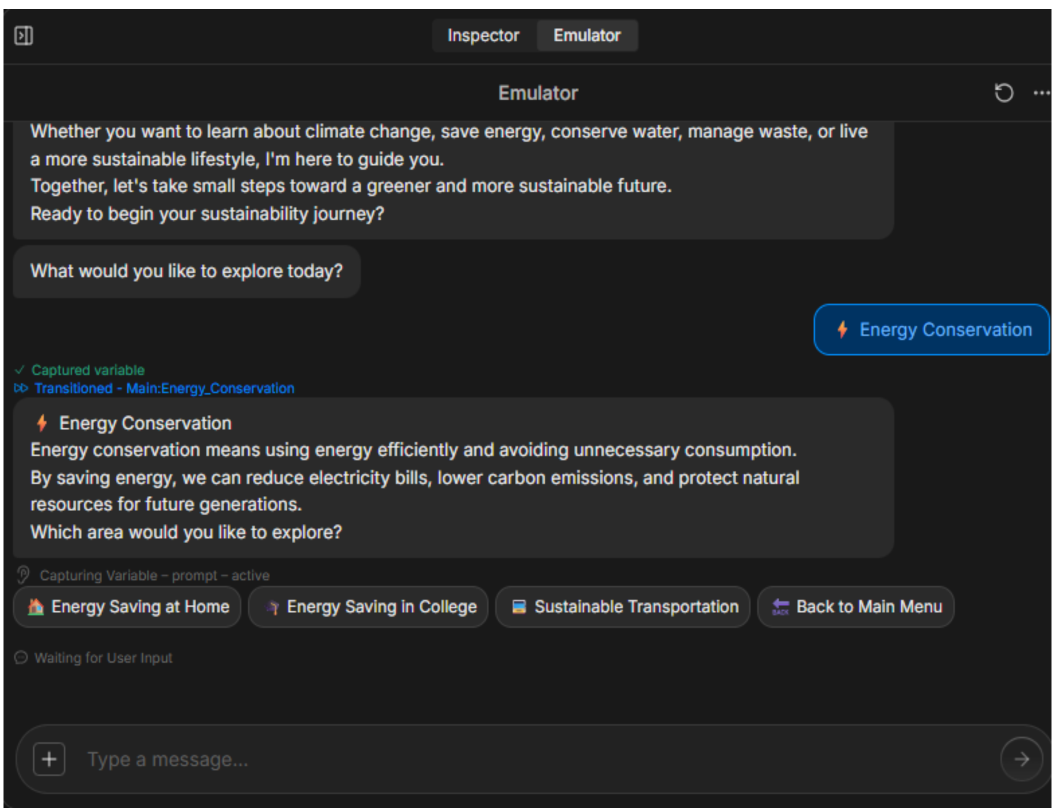


➢ Energy Saving



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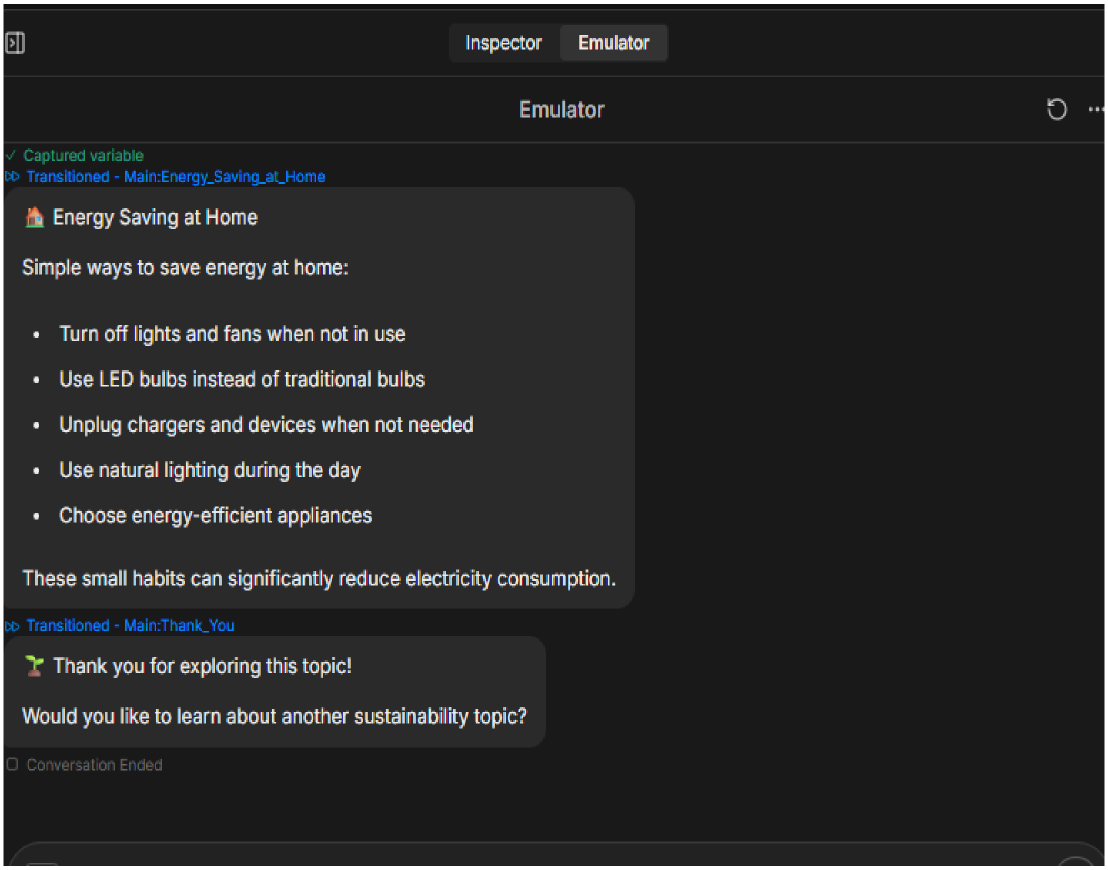
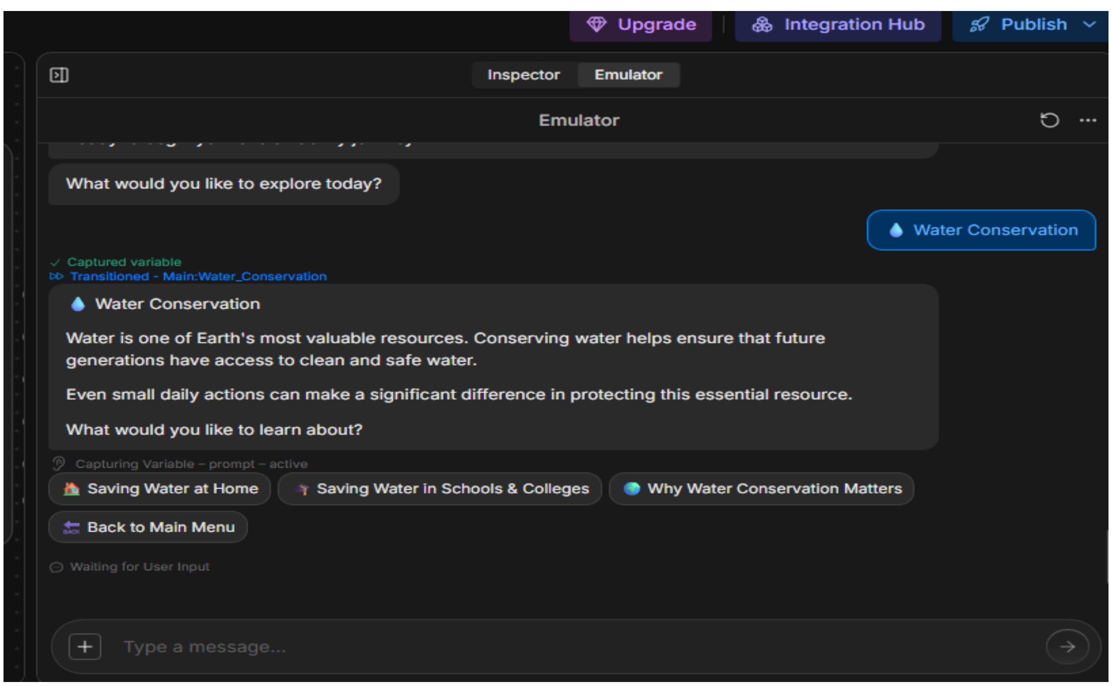


➢ <u>Water Conservation</u>



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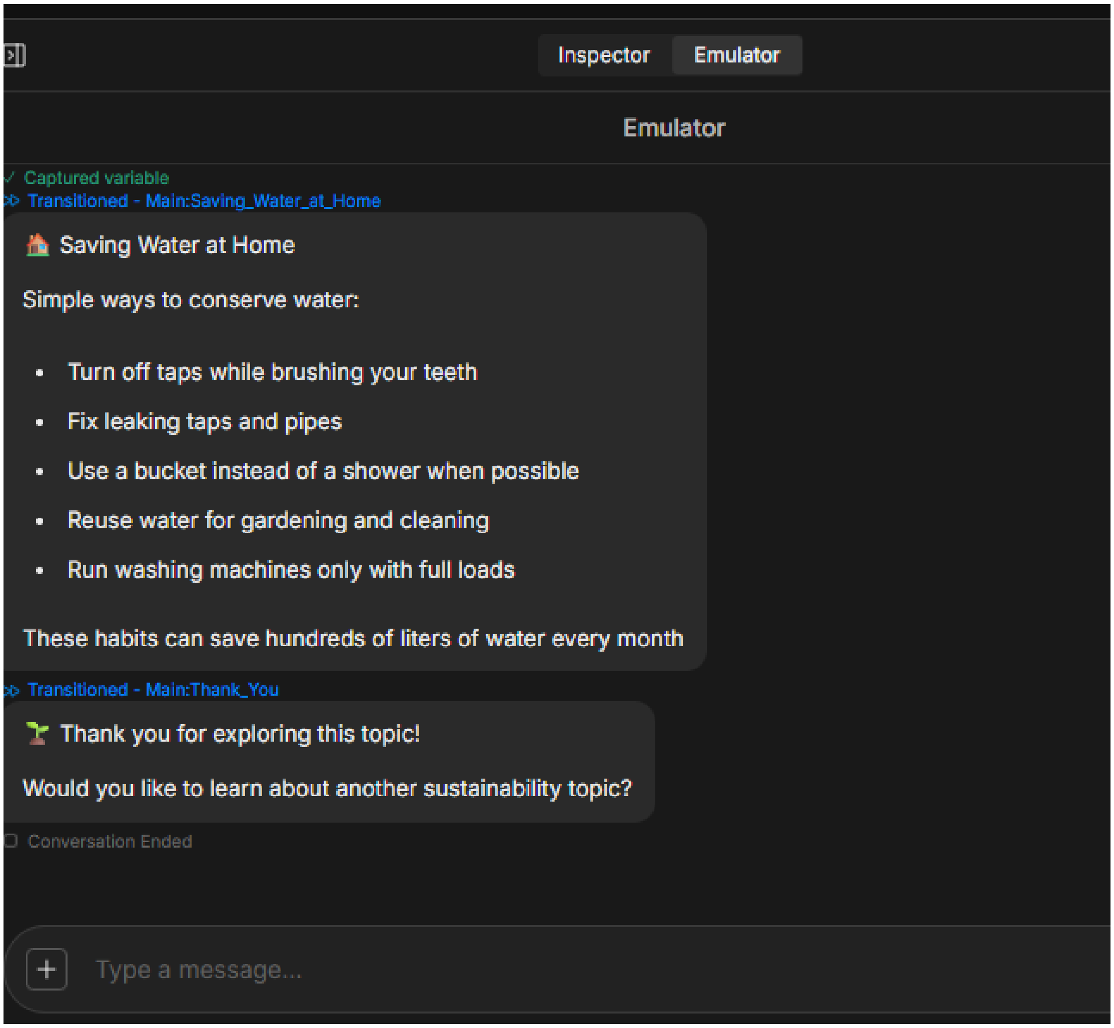
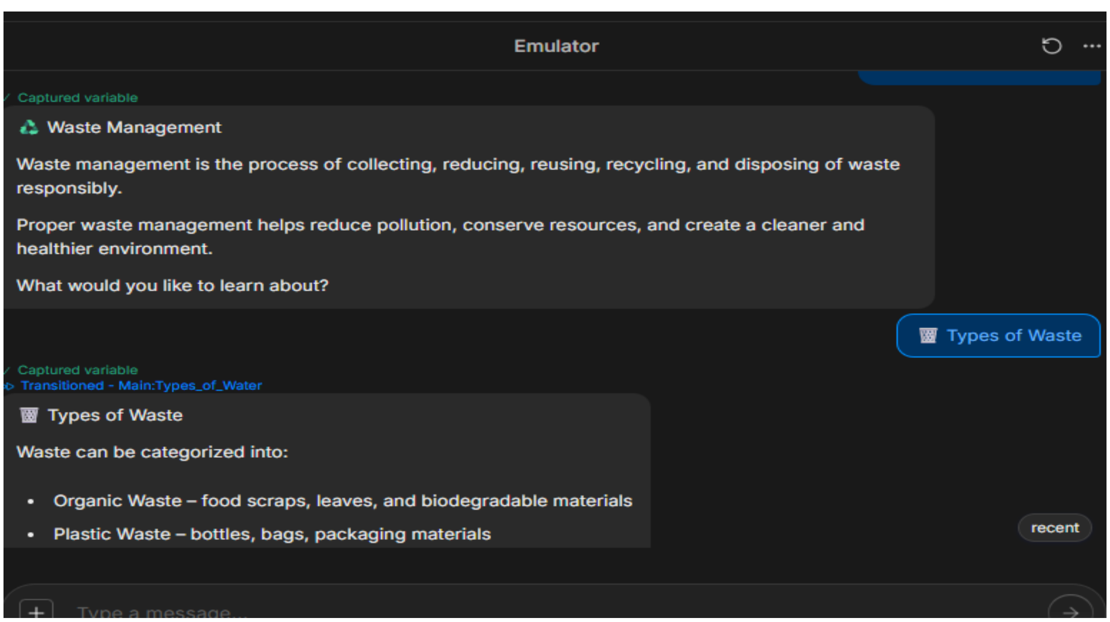


➢ Waste Management



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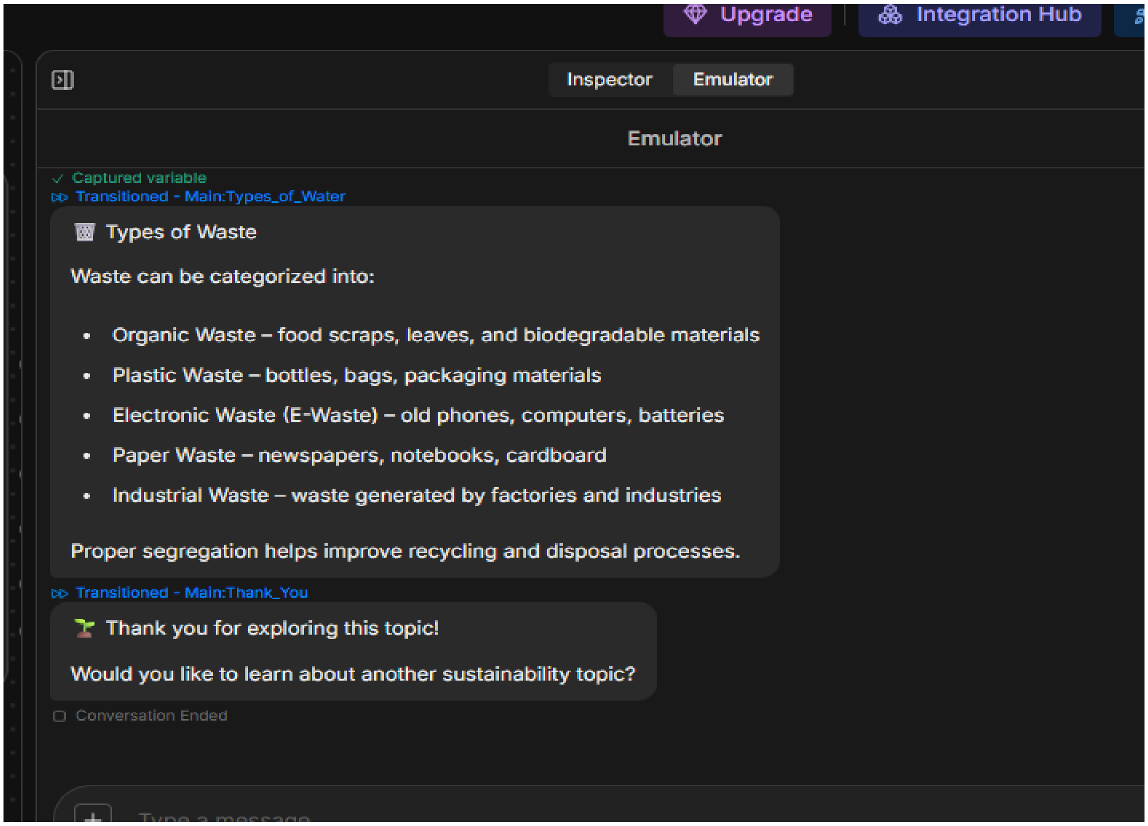
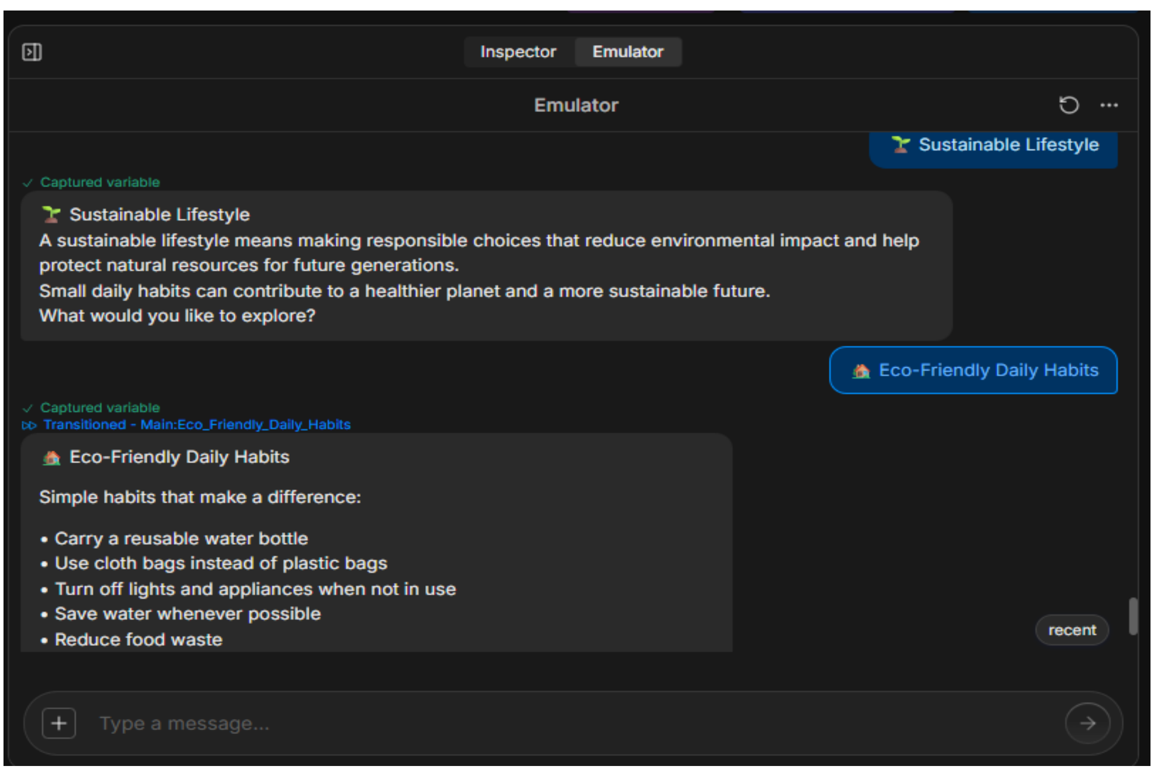


➢ <u>**Sustainable Lifestyle**</u>



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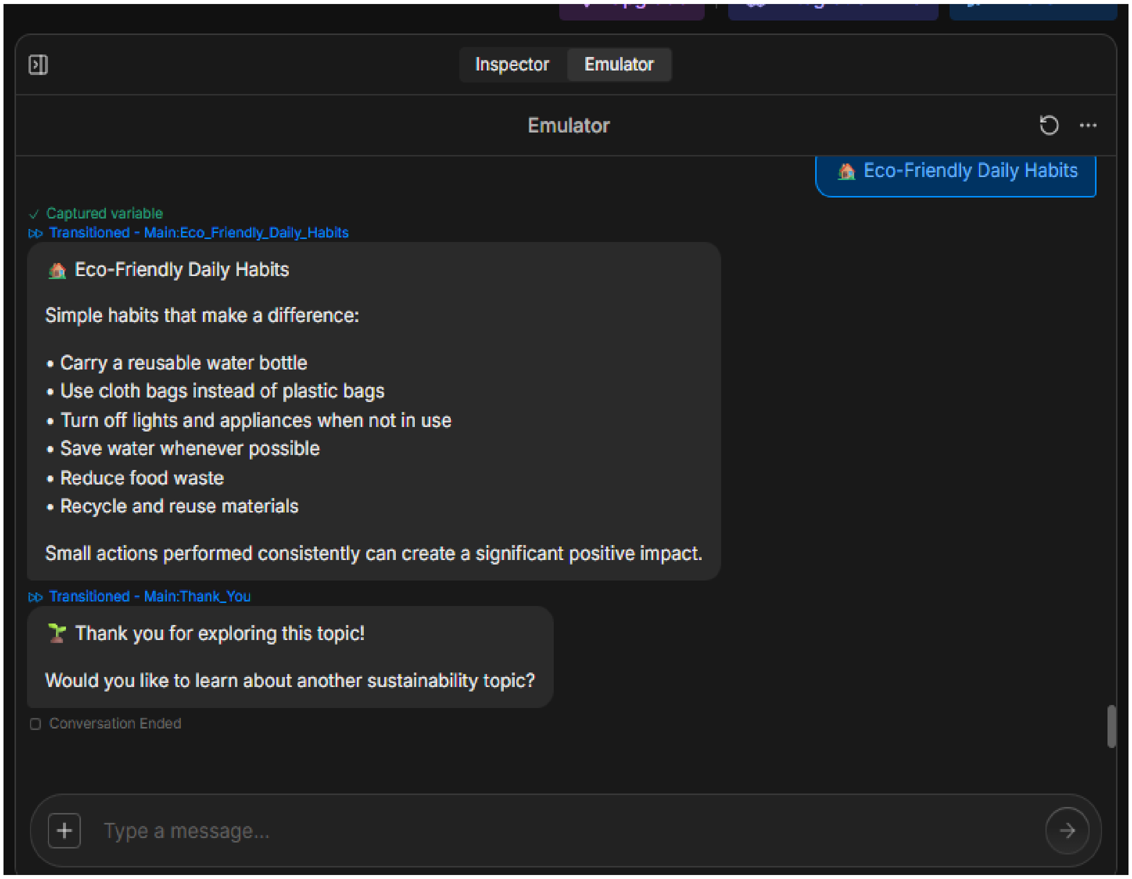

Project Links

Live Chatbot Link

https://cdn.botpress.cloud/webchat/v3.6/shareable.html?configUrl=https://files.bpcontent.cloud/2026/06/22/09/20260622095708-4C306M43.json

GitHub Repository

https://github.com/radhikamahajan705-afk/TerraSense_AI_Sustainability_Assistant

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Conclusion & Summary

TerraSense AI is a sustainability assistant chatbot developed to support SDG 13: Climate Action.

The chatbot helps users learn about climate change, energy conservation, water conservation, waste management, and sustainable lifestyles through an interactive conversational interface.

The project demonstrates how conversational AI can be used to promote environmental awareness and encourage responsible behavior. By providing practical sustainability guidance, TerraSense AI contributes to building a greener and more environmentally conscious society.